

Loose Puzzle Pieces II: Cosmology

Redshift, scales and the boundary of the cosmic sector

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Loose Puzzle Pieces II: Cosmology

Abstract

Doc. 390 placed the building blocks of particle physics that were found in the literature afterwards and fit into the picture of FFGFT. This document does the same for cosmology. It goes through eight building blocks: Zwicky's tired light, the gravitation theory of Hoyle and Narlikar and Wetterich's universe without expansion, Einstein's static universe and the steady-state theory, Dirac's large numbers, the Hubble length, Milgrom's acceleration scale, the problem of vacuum energy, and the cosmic background radiation. The order holds here too: the building blocks were not the foundation of FFGFT; they fit in afterwards. Unlike in particle physics, however, the cosmic sector lies outside the claim of derivation: H_0 is calibrated to the measurement, the exponent 10 is a setting, and the Standard Model does not derive this sector either. The puzzle pieces therefore show readings, not derivations: what the observations of cosmology carry can be read with $\hat{T} \cdot m = 1$ without expansion, and several places in the literature have already shown this. Two building blocks, vacuum energy and background radiation, do not fit so far or fit only as a candidate, and are named as the boundary.

Status markers

[SETZUNG]	Motivated input, not a derivation.
[B]	Algebraically proved (verification script, exact arithmetic).
[K]	Numerically confirmed.
[S]	Structurally plausible, identification open.
[E]	Established mathematics or physics, adopted.
[Q]	Taken from the literature, with source.
[X]	Checked and negative.

The markers give the status *in the corpus*; the document derives nothing new but places things in context. Verification script: `pruef_391_kosmologie.py`.

1 Order and boundary

On the order. As in Doc. 390, none of the following building blocks was a starting point of FFGFT. The theory was developed from $\tilde{T} \cdot m = 1$ and the geometry of the carrier; the literature references were found only afterwards. Where it says that a building block fits in at some place in the corpus, this is agreement in hindsight, not origin.

On the boundary. The cosmic sector is deliberately left out of the corpus. The Standard Model does not derive H_0 and Λ either, so a comparison would be asymmetric (P39 in Doc. 190; Doc. 365). The form of the relation between the Hubble length and the Planck length is fixed as a dimensionless ξ ratio (P20), the exponent 10 in $H_0 \propto \xi^{10}$ is a setting **[SETZUNG]**, and the scale is calibrated to the measurement of H_0 (R137 in Doc. 190). What this document places are therefore readings of the observations, not predictions of their values.

2 Redshift without expansion

Zwicky's tired light (1929)

What it contained. Shortly after Hubble's discovery, Zwicky proposed that the redshift of distant galaxies arises not from expansion but from light losing energy along the way **[Q]** [1]. This was the first static interpretation of the Hubble relation.

What was missing. Tired light changes only the photon. Distant clocks, such as the light curves of supernovae, would then have to run at their original rate. What is observed, however, is a stretching by $(1 + z)$, and the idea failed on that.

Where it fits in the corpus. Because of $\tilde{T} \cdot m = 1$, wavelength and clock rate are booked together. From $1 + z = e^{\xi_0 x}$ the time dilation $(1 + z)$ follows for all z **[B]** (R128 in Doc. 190). Classical tired light fails because it changes only the photon side; the mass reading cannot make this mistake structurally (Doc. 312). The redshift is achromatic, and the scale of the path lengthening is calibrated via H_0 (K6 in Doc. 190).

Hoyle–Narlikar (1964) and Wetterich's universe without expansion (2013)

What it contained. Hoyle and Narlikar formulated a theory of gravitation in which particle masses arise from the distribution of matter in the cosmos **[Q]** [2]. In 2013 Wetterich showed that an expanding universe with constant masses turns, through a field redefinition, exactly

into a static universe with growing particle masses **[Q]** [3]. Only dimensionless ratios are observable.

What was missing. A reason why the running of masses should be the natural reading and not merely an equivalent rewriting.

Where it fits in the corpus. Doc. 312 lists the equivalence of expansion and running masses as **[K]** and thereby anchors the degeneracy from Doc. 267: redshift, luminosity distance, baryon acoustic oscillations and background temperature do not distinguish the two readings. In FFGFT the running of masses is not an additional assumption, because mass and clock rate are the same quantity through $\tilde{T} \cdot m = 1$ anyway. Doc. 312 records the boundary: the equivalence shows that the static reading is possible and observationally equal, not that it is right.

Einstein's static universe (1917) and the steady-state theory (1948)

What it contained. In 1917 Einstein built the first relativistic world model as a static, spatially closed cosmos **[Q]** [4]. In 1948 Bondi, Gold and Hoyle formulated a universe without a beginning that looks the same on large scales at all times **[Q]** [5, 6].

What was missing. Einstein's model needed a specially introduced cosmological constant and was unstable. The steady-state theory needed a continuous creation of matter, and the discovery of the background radiation in 1965 was taken as its end.

Where it fits in the corpus. FFGFT lives on a carrier without boundary, $\partial T^4 = \emptyset$ (Doc. 312). Doc. 313 works through the question of the beginning on the compact time cycle: the observable history fills half a cycle, and the "beginning" is the antipode of the cycle, not an event **[K]** with the calibration from P20. Neither a cosmological constant nor creation of matter is needed for this. According to Doc. 313 it remains open that an exact periodicity of the cycle is not dynamically generic and has to be set as a boundary condition.

3 Scales

Dirac's large numbers (1937)

What it contained. Dirac noticed that the ratio of the electric to the gravitational force between electron and proton and the age of the universe in atomic units of time are both about 10^{39} , and suspected a connection behind this **[Q]** [7].

What was missing. A mechanism. Dirac's conclusion that the gravitational constant must decrease with time is excluded by measurements.

Where it fits in the corpus. In the corpus the relation between Hubble length and Planck length is a dimensionless ξ ratio (P20); the exponent is set, and the scale is calibrated to H_0 . P39 records that no framework derives this relation from first principles. Dirac's observation thus appears as a ratio of two scales of the same geometry, not as a constant running in time. Dirac's name does not yet occur in the corpus; the placement is a reading **[S]**.

The Hubble length as maximal scale

What it contained. In 1929 Hubble found the linear relation between distance and redshift **[Q]** [8]. Since then the length c/H_0 has been read as the horizon of an expanding universe.

What was missing. A reading of this length without expansion.

Where it fits in the corpus. Doc. 182 and Doc. 365 read $R_H = c/H_0$ as the geometric maximal scale of the compact carrier, not as an expansion horizon. Its value is calibrated to H_0 .

Milgrom's acceleration scale (1983)

What it contained. Milgrom explained the flat rotation curves of galaxies without dark matter through an acceleration scale a_0 below which Newton's law deviates **[Q]** [9]. He already noticed that a_0 is of the order of cH_0 .

What was missing. A reason for this closeness to the cosmic scale, and a justification of the interpolation formula.

Where it fits in the corpus. In the corpus $a_0 = cH_0/(2\pi)$ **[K]** (Doc. 365); the same exponent 10 that carries H_0 also reproduces a_0 , which would not be expected from a pure H_0 fit (Doc. 309, cross-anchoring). Doc. 308 obtains the form $a = \sqrt{a_N^2 + a_N a_0}$ with $a_0 = 2a_{bg}$ from the Unruh coupling **[K]/[S]**. The book "Dark Matter – an Illusion" works out the rotation curves. The exponent remains set.

4 Where it does not fit yet, or only as a candidate

Vacuum energy and the cosmological constant

What it contained. The zero-point energy of quantum fields is measured in the Casimir effect. As a source of the cosmological constant, its naive estimate lies many orders of magnitude above the observed value; Weinberg called this the cosmological constant problem **[Q]** [10].

Status in the corpus. The connection between the Casimir and background energy densities at the length L_ξ is an identity, not evidence from measurement **[S]** (R136 in Doc. 190). The Λ formulas of older documents do not hit the observed value **[X]** (R137). In the static reading, Λ is an artefact of the reading without a referent of its own (Doc. 308) and is not set (Doc. 312); there is no derivation of the observed value. This building block does not fit in so far.

The cosmic background radiation (1965)

What it contained. In 1965 Penzias and Wilson found an isotropic radiation of about 3 kelvin **[Q]** [11]. Its spectrum and its anisotropies are among the most precise data in cosmology.

Status in the corpus. The relations for T_{CMB} from ξ give a pure number that agrees with the measured value only in the unit eV; they count as a unit-dependent agreement, and the question is open (R137, R70). The relation $T(z)$ in Doc. 039 is excluded by measurements **[X]**. The factor 3 in front of the peak sequence is a setting, and the peak selection as a whole is open (R139). In the basic form, Doc. 388 sets up the candidate $T_{CMB}/m_e = (8\pi)^{1/4} \xi^{5/2}$, which hits the FIRAS value **[K]**; the prefactor is motivated but not justified **[S]**, and the random test there shows that chance is not excluded. If the candidate holds, it connects temperature and Hubble scale through $T_{CMB}^4 = 16 H_0 m_e^3$ **[B]** (Doc. 388), with the same set exponent 10. The temperature of the radiation is moreover among the quantities that do not resolve the degeneracy from Doc. 267. This building block thus fits in only as a candidate, not as a secured place.

5 What the pieces show

In particle physics (Doc. 390) the building blocks fitted in at places where the corpus derives something. In cosmology they fit in at places where the corpus *reads* something. Zwicky, Hoyle and Narlikar, Wetterich, Einstein and the steady-state theory each supplied a piece of a static cosmology; each piece failed on its own at a different point, at time dilation,

at the missing justification of running masses, at instability or at creation of matter. With $\tilde{T} \cdot m = 1$ these points are no longer separate problems, because wavelength, clock rate and mass are the same booking.

Dirac and Milgrom each noticed a striking closeness between microscopic and cosmic scales. In the corpus both are ratios of the same geometry, but tied to a set exponent and a calibrated scale. They are therefore cross-anchorings, not evidence.

Vacuum energy and background radiation do not fit so far, or only as a candidate. This is not concealed: the formulas concerned are listed in the register as **[X]** or as open, and the temperature candidate from Doc. 388 as **[S]**. For the cosmic sector the same therefore holds as for scheme-dependent quantities in particle physics: no claim of derivation is made where the quantities depend on the measurement or on a convention.

6 Overview

Building block	Year	What was missing	In the corpus
Zwicky, tired light	1929	no time dilation	R128, K6; Doc. 312
Hoyle–Narlikar, Wetterich	1964, 2013	running masses only equivalent, not justified	Doc. 312, 267
Einstein universe, steady state	1917, 1948	instability, creation of matter	Doc. 313, 312
Dirac’s large numbers	1937	no mechanism	P20, P39 [S]
Hubble length	1929	read only as expansion horizon	Doc. 182, 365
Milgrom’s a_0	1983	closeness to cH_0 unexplained	Doc. 365, 309, 308
Vacuum energy, Λ	1989	does not fit so far	R136, R137
Background radiation	1965	only a candidate, chance not excluded	Doc. 388; R137, R139, R70

7 Conclusion

In cosmology, too, the puzzle pieces were in the literature long before FFGFT, and here too they were not its foundation. Six of the eight building blocks fit in afterwards, as readings of the observations without expansion and as cross-anchorings between microscopic and cosmic scales. Two do not fit so far or fit only as a candidate. The difference from particle physics must be stated honestly: the cosmic sector is calibrated and left out; here the building blocks show that the theory can read cosmology, not that it derives its numbers.

Verification script: 2/python/Dok391_Skripte/pruef_391_kosmologie.py — 51/51.

Tool and method

Claude (Anthropic) was used as a tool for: compiling the references in the corpus, writing and running the verification script, and working out the $\text{\LaTeX}$ sources according to the author’s specifications. The verification script backs every corpus reference in this document with a text check in the named chapter file; if a check fails, the reference is not used.

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